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The promising future of game development

Today's game engines offer much more to game developers than ever before. Taking a game from idea to final product has always been a daunting task, but with today's development platforms, it is easier than ever. Creators no longer need to write or develop a game from the scratch. Ideas are much more easily implemented with the use of a modern games engine, designed to reduce as much friction as possible in the creation process. In engines, the basic core architecture and programming is largely provided. A developer only needs to tweak them as their own requirement, adding in layers of their own creation to bring their vision to life.

In comparison to the early days, game engines have become more user-friendly and continuously evolve with time. Development is becoming democratized, both due to the increasing ease of use and the affordability. Thanks to their ability to provide creation tools across a variety of industries, the term "game engine" is really becoming a misnomer. The Unity real-time 3D development platform can be used in just about any field, developing architectural simulations, medical training aids, automotive renderings and more.

Today's game engines are also more diverse in terms of platform support, allowing developers to target the likes of smartphones, desktops, home consoles and AR/VR devices.

Oclulus Go is an untethered VR experience that brings mobility and accessibility to VR, something that it has desperately needed since its introduction to the mass markets

The future is bright in the world of Augmented and Virtual Reality: AR is here and VR is promising greater growth for 2018. AR is no longer the future, it's here and will be available on more than 1 billion devices in the marketplace by the end of 2018.

The global virtual reality (VR) market was valued at approximately USD 2.02 billion in 2016 and is expected to reach approximately USD 26.89 billion by 2022, growing at a CAGR of around 54.01% between 2017 and 2022, according to Zion Market Research's report. In the last 12 months to the end of Q1 2018, augmented and virtual reality startups raised over $3.6 billion from venture capitalists and corporates, setting a new record. Over three-quarters of a billion dollars was invested in the first three months of this year alone. Additionally, with the proliferation of untethered all-in-one VR devices from Google and Oculus are leading us to greater consumer awareness.

Which brings us to another avenue – Artificial Intelligence. Research in artificial intelligence is unfolding new avenues in the continuously evolving relationship between human game developers and their algorithmically-intelligent tools. Entire 2D/ 3D games can be brought to life from the beginning with the help of artificial intelligence. AI will be able to provide creative support, where intelligent design tools help people to quickly generate layouts, characters, themes, environments and so on.

In Unity's real-time 3D development platform, our creators are using Machine Learning Agents to train behaviors of non-player characters and other supporting elements. This breakthrough open source AI toolkit enables machine learning developers and researchers to train agents in realistic, complex scenarios with decreased technical barriers than they could otherwise.

For game design, artificial intelligence is a key to tackle a wide range of gaming problems. In a world where most current genres and core gameplay mechanics have been around for years, refined again and again to a sharp point, AI has the potential to bring something new to the table. d